PAPER ID-311384

Subject Code: BCE057

Roll No: | | | | | | | | | | | | | |

BTECH
(SEM V) THEORY EXAMINATION 2025-26
AIR AND NOISE POLLUTION CONTROL

TIME: 3 HRS **M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 07 = 14**

Q No.	Question	CO	Level
a.	Describe the global implications of air pollution with respect to climate change and ozone layer depletion.	1	
b.	What is indoor air pollution? Explain the major indoor air pollutants and their sources.	1	
c.	What is temperature lapse rate? Discuss its role in determining atmospheric stability.	2	
d.	Explain the concept of atmospheric turbulence and its importance in air pollution dispersion.	2	
e.	What is the Air Quality Index (AQI)? Explain its significance in public health communication.	3	
f.	Discuss ambient air quality and explain the need for ambient air quality standards.	3	
g.	Write a short note on the working of a catalytic converter used in automobiles.	4	

SECTION B

2. Attempt any *three* of the following: **07 x 3 = 21**

a.	Discuss the classification of air pollutants and explain the formation, sources, and environmental effects of oxides of sulphur and oxides of nitrogen.	1	
b.	Explain the process of dispersion of air pollutants in the atmosphere, highlighting the roles of advection, diffusion, and turbulence.	2	
c.	Explain the design, working principle, advantages, and limitations of electrostatic precipitators (ESP) used in air pollution control.	3	
d.	Compare Euro-I, Euro-II, and Euro-III emission standards and discuss their adoption and implementation in the Indian context.	4	
e.	Explain the basics of acoustics and discuss the specification of sound, including sound power, intensity, and sound pressure levels.	5	

SECTION C

3. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Discuss the global impacts of air pollution, including acid rain, global warming, and trans-boundary air pollution, with suitable illustrations.	1	
b.	Explain the formation and effects of photochemical oxidants, highlighting the role of hydrocarbons and nitrogen oxides in photochemical smog.	1	

4. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Describe the Gaussian Plume Model in detail, including its assumptions, governing equation, and applications in air quality prediction.	2	

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b.	Discuss the meteorological factors affecting air pollution dispersion, with special reference to temperature lapse rate, wind speed, and atmospheric stability.	2	

5. Attempt any *one* part of the following: 07 x 1 = 07

a.	Explain in detail the Air Quality Index (AQI), including its calculation, categories, and health impacts.	3	
b.	Compare gravitational settling chambers, wet collectors, and cyclone separators for the control of particulate air pollutants with respect to efficiency, applications, and limitations.	3	

6. Attempt any *one* part of the following: 07 x 1 = 07

a.	Describe automotive emission control strategies, explaining the design and functioning of catalytic converters in reducing exhaust emissions.	4	
b.	Explain in detail the control technologies for sulphur oxides, nitrogen oxides, carbon monoxide, and hydrocarbons from stationary and mobile sources.	4	

7. Attempt any *one* part of the following: 07 x 1 = 07

a.	Describe various special noise environments such as infrasound, ultrasound, impulsive noise, and sonic boom, and discuss their effects.	5	
b.	Discuss noise instrumentation and monitoring procedures and explain different noise control methods at source, path, and receiver.	5	

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